

Lab 3 Report – Unsupervised Learning

Basic principle of PCA

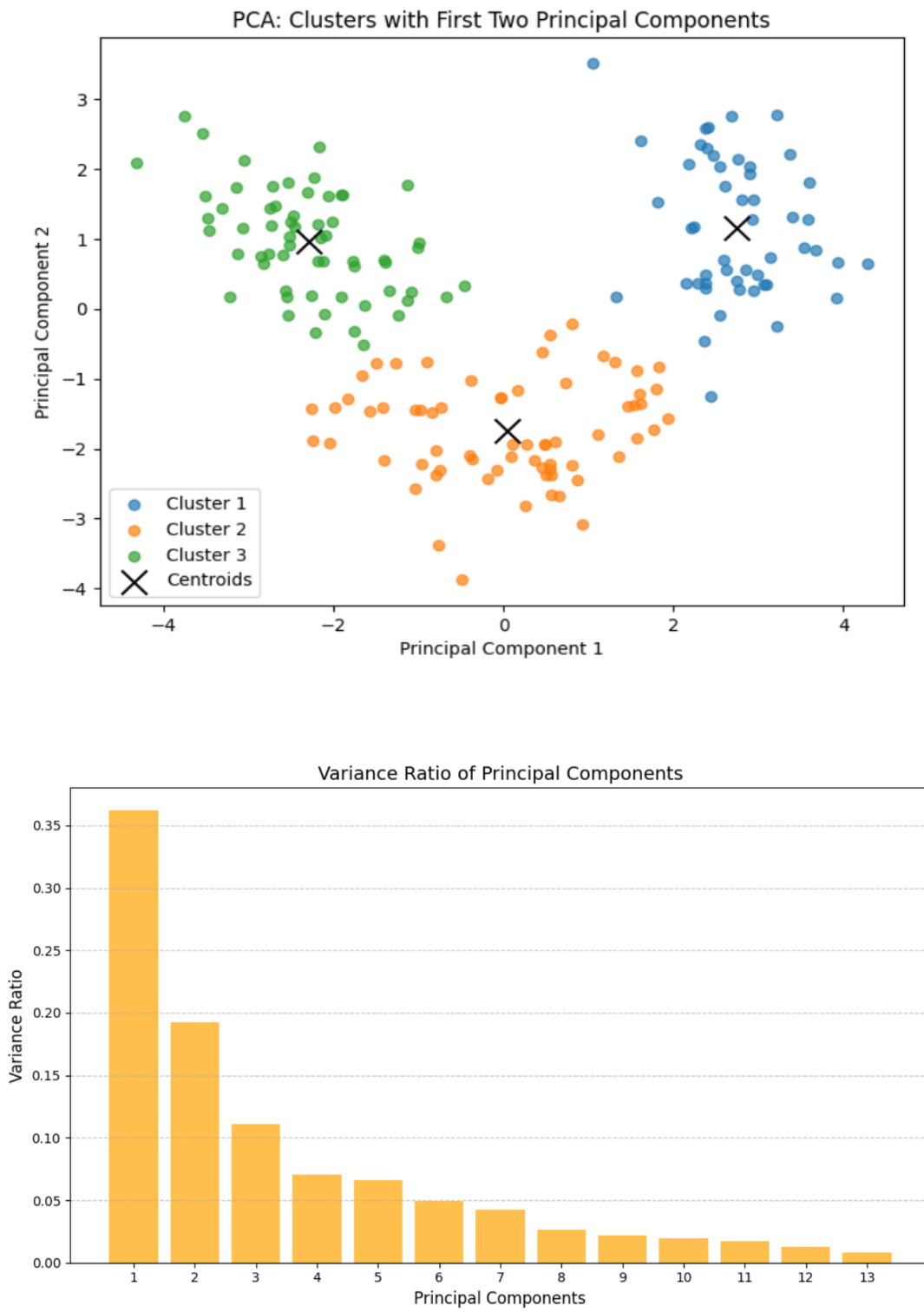
Principal component analysis (PCA) is a dimension reduction technique used in machine learning to transform and simplify data while preserving as much information about its variability as possible. It begins by standardizing the data beforehand to make sure each feature contributes equally to the initial data analysis. By computing a covariance matrix of the data set, one obtains the covariance between features, each variance, and how they are related. Eigenvalues and eigenvectors are calculated using these values of covariance and are used to create the new features. By projecting the eigenvectors whose respective directions are in the direction of most variance of data, the new features cover the variability of the original data and are orthogonal to one another. A principal component is a linear combination of the original features, and they are ranked in order of their eigenvalue, with higher magnitude of eigenvalue meaning larger variance covered of the original data.

K-means algorithm

K-means algorithm is an unsupervised machine learning algorithm used to partition data into k clusters based on their proximity to cluster centroids. The algorithm begins by placing k number of centroids at random coordinates. Data points are then all assigned to the nearest centroids, forming clusters of data points. After all data points have been assigned, the centroids are recalculated by computing the mean value of all the points within each cluster. It will then again assign all of the data points to the new nearest centroids. This process will either continue until the maximum number of iterations is reached or when centroids no longer move, which means clustering has converged.

The main advantages of k-mean clustering are that it is simple to implement and can be computed efficiently. However, the downside is that it requires a predefined number of clusters k and is sensitive to the initial placement of the centroids, which can lead to local minima.

Figure



Table

2 principal components

Cluster	X-Coordinates	Y-Coordinates	Data points
Centroid 1	2.738	1.165	50
Centroid 2	0.041	-1.743	67
Centroid 3	-2.289	0.959	61

5 principal components

Cluster	X	Y	Z	...	...	Data points
Centroid 1	2.720	1.126	-0.239	-0.062	0.074	51
Centroid 2	0.037	-1.772	0.186	-0.80	0.071	65
Centroid 3	-2.276	0.932	0.002	0.135	0.135	62

8 principal components

Cluster	X	Y	Z	...	...	...	...	...	Data points
Centroid 1	2.720	1.126	-0.239	-0.062	0.074	0.010	0.066	0.007	51
Centroid 2	0.067	-1.762	0.310	-0.118	-0.031	-0.112	0.066	-0.028	64
Centroid 3	-2.270	0.879	-0.122	0.170	-0.029	0.194	-0.116	0.023	63